

Satellite Communications

ECE 4644 / 5610 Spring 2026

Week #2 – First Lecture

Jan 26 - 30, 2026

Instructor: J. Michael Ruohoniemi

ECE 4644 / 5610: Preliminaries

- Call me:

Dr. Ruohoniemi (Row-hawn'-ah-me)

or

Professor

or

Dr. 'R'

ECE 4644 / 5610: Preliminaries

- It is easiest to contact me by email at
mikeruo@vt.edu
- Remember to include 'ECE 4644' or
'ECE 5610' in the subject heading of your
emails

1st Lecture Topics

- Introduction to satellite communications
- Brief history of satellite communications
- Course details
- Satellite orbits: GEO, MEO, LEO

? Why Satellite Communications ?

What is Satellite Communication?

- Humans have a basic need to communicate (news, information, stories, gossip, romantic interest...)
- Sometimes they want to communicate over long distances
- Prior to the advent of telecommunications, long-distance communication was slow and arduous and required a physical messenger, e.g., the Pony Express

What is Satellite Communication?

- Since the advent of the telegraph various technologies have been developed to enable rapid long-distance communications
- Satellite Communication uses artificial satellites to create communications links between widely separated points
- Satcom presents distinct advantages, especially for communicating over wide areas

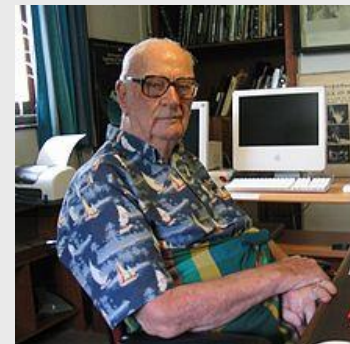
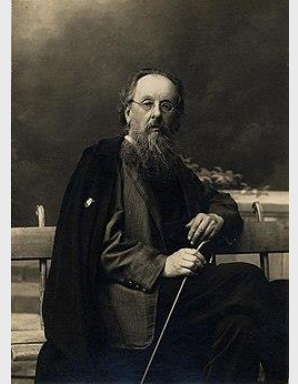
Evolution of Telecommunications

- Copper wires
- Telegraph 1840
- Telephone 1880
- Radio
- Broadcasting 1920
- Line of sight microwave links 1950
- Satellite Communications 1965
- Optical fiber
- High capacity, long distances 1985

Satcom Telecommunication Milestones

Date Achievement

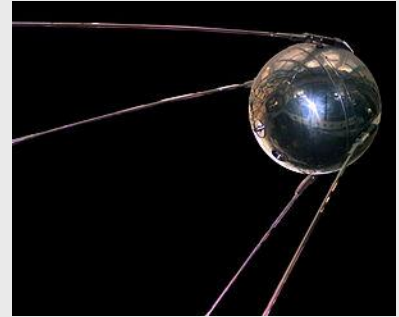
- 1897 Russian scientist Konstantin Tsiolkovsky develops theory of rocket propulsion
- 1928 Slovenian engineer Herman Potočnik proposes earth observation with orbiting satellites and a design for a space station
- 1945 Arthur C. Clarke proposes communications with satellites in geostationary orbit



Satcom Telecommunication Milestones

Date Achievement

- Oct 1957 USSR launches Sputnik



- Jan 1958 US launches Explorer I



Satcom Telecommunication Milestones

Date

Achievement

- Jan 1959 Project SCORE:
first communications satellite
- May 1960 Project ECHO:
passive communications satellite
- 1962-3 Telstar satellites: first transatlantic
satellite links
- 1964 Intergovernmental organization
Intelsat formed



Satcom Telecommunication Milestones

Date	Achievement
• 1965	First commercial GEO satellite Intelsat I (Early Bird)
• 1968	Intelsat III satellites form global communications network
• 1974	Domestic satellite comm begins
• 1978	First GPS (Navstar) satellite
• 1985	VSAT systems started
• 1993	First DirecTV satellite launched
• 2001	XM satellite radio begins national service



SatCom versus Optical Fiber

- Optical fiber offers much higher rates of data transmission and lower cost but works only for *point-to-point communications*
- Satellite communications work over oceans, the poles, and as you move around
- Satcom is preferred for *broadcasting* and for communications to remote settings (most of the world)

SatCom today

- There are currently more than 15,000 active satellites in orbit versus $< 1,500$ in 2016
- All satellites must be equipped with some communications capability
- Satcom refers primarily to satellites whose purpose is to facilitate communications between users separated by large distances
- Satcom accounts for much more than half of all working satellites

SatCom today

- Primary uses for satcom are telephony, DBS-TV, video distribution, data transmission, and internet access
- Commercial viability is a major concern for the design and operation of satcom systems

SatCom and Global Broadband

- The promise of *global broadband* is driving the rapid increase in the number of satellites
- Elon Musk's entry into the field is SpaceX's *Starlink*
- The Starlink constellation consists of > 9,400 satellites
- There were >100 Falcon 9 launches and 5 Starship launches in 2025



60 Starlink (SpaceX) satellites stacked together before deployment

Global satellite system *OneWeb* (1/4)

- Commercial startup *OneWeb* also envisioned a satellite constellation to provide world-wide high-speed internet access
- Individual satellites cost between \$500k and \$1M
- First large batch of 34 satellites was launched in February 2020



Global satellite system *OneWeb* (2/4)

- March 2020 – OneWeb files for bankruptcy with 72 satellites in orbit
- November 2020 – OneWeb exits bankruptcy due in part to investment by the UK government
- January 2021 – OneWeb files a request with the FCC for permission to fly 6,372 satellites
- Major costs, major risks, major rewards...

Global satellite system *OneWeb* (3/4)

- March 2022 – OneWeb abandons launching from Russia, reaches agreements with SpaceX and the Indian Space Research Organization (ISRO)



Stack of 36
OneWeb
satellites

Global satellite system *OneWeb* (4/4)

- March 2023 – ISRO launches the last tranche of OneWeb satellites from the Sriharikota spaceport to complete the initial 618-satellite constellation, sufficient to provide global coverage (now 648 satellites)
- September 2023 – France's Eutelsat merges with OneWeb to create 'Eutelsat OneWeb'
- Now offering global broadband services for industrial and government customers (not individual consumers)

Global satellite systems: Summary

- Large constellations of satellites flying at low altitude are upending the traditional satcom paradigm based on flying a few large, expensive satellites at high altitude
- Enabled by great reductions in launch cost
- Companies include Starlink, OneWeb, and Amazon's Kuiper (Ki'-per), Qianfan ('Thousand Sails', China), IRIS (EU)
- 'Legacy' satcom companies are adapting, e.g., Eutelsat

First flight of New Glenn (Blue Origin) rocket

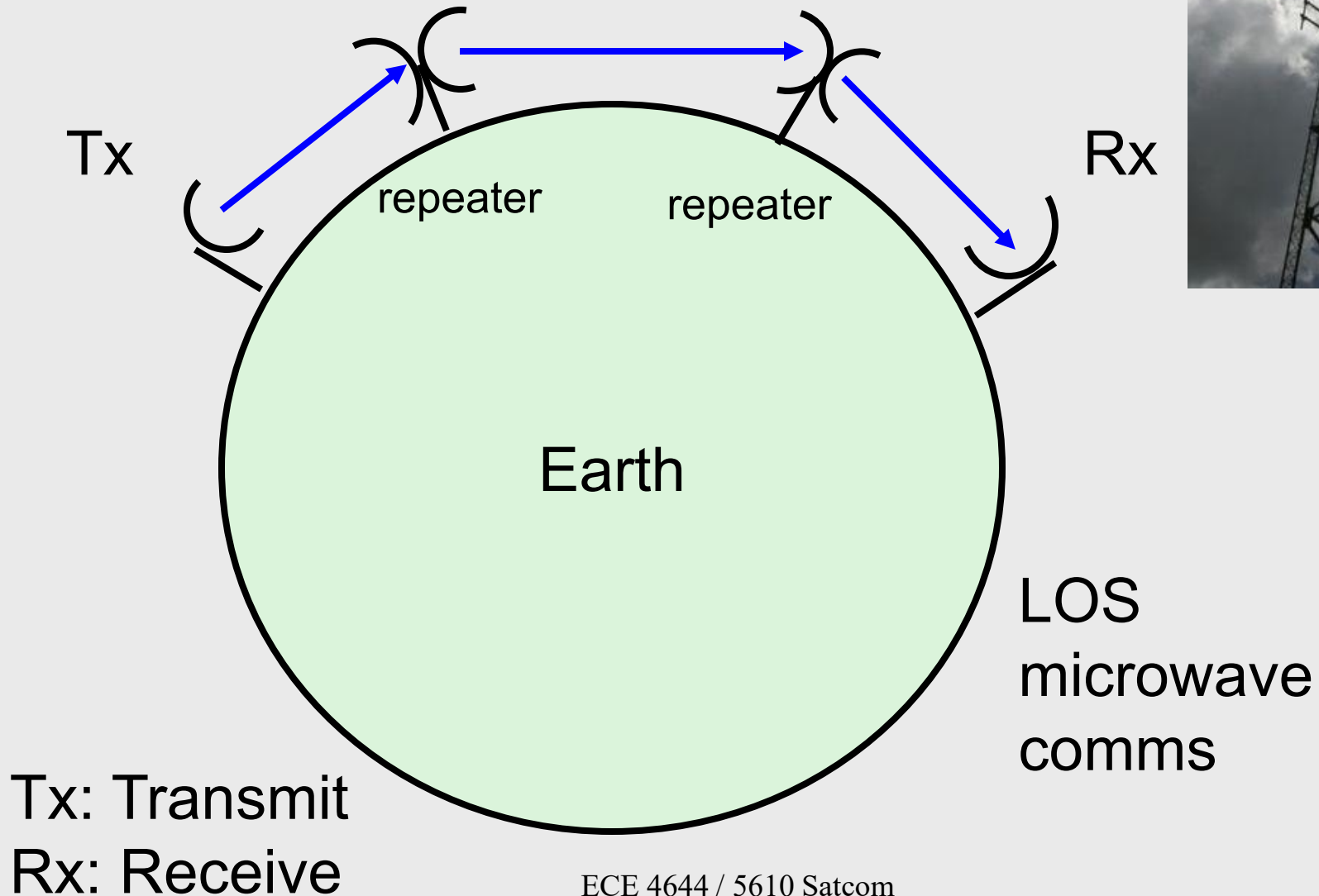


Launch from Cape Canaveral on January 16, 2025

Long-Distance Radio Communications

- Radio signal can be applied to long distance communications
- Radio waves travel in straight lines
- The body of the Earth represents an obstacle to radio communications
- Maximum line of sight distance on flat land is about 80 km
- Consider three possibilities

Microwave communication link



Long-Distance Radio Communications (1/3)

- *Repeaters* mounted on towers can be used to get round the curved surface of Earth (a term first used with telegraphy)
- A repeater receives a signal, amplifies and retransmits it in a new direction
- Basis of terrestrial microwave links

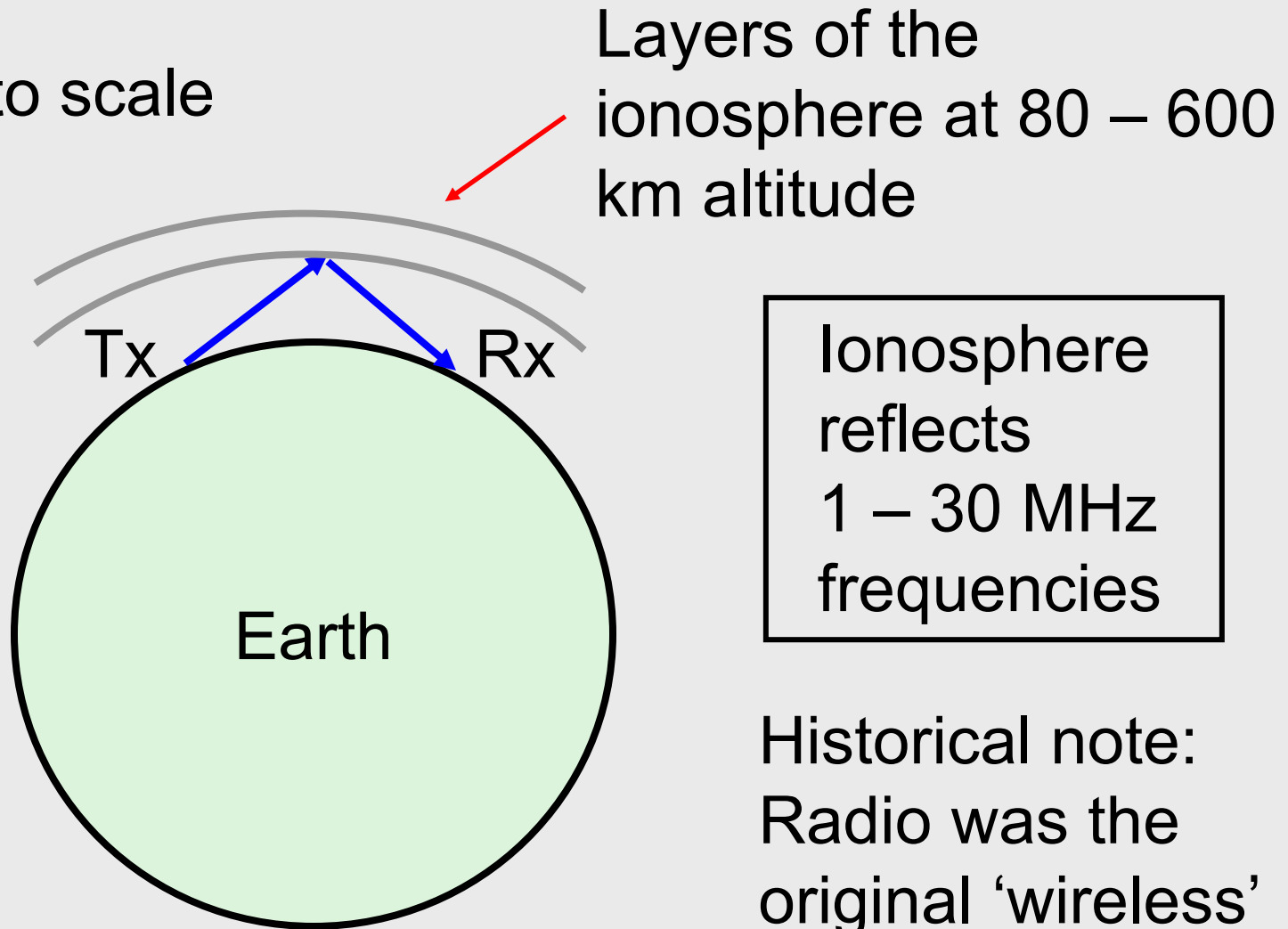
Long-Distance Radio Communications (2/3)

- We can reflect radio waves from the ionosphere to reach long distances
- Useful over a limited range of frequencies: 1 – 30 MHz (MF / HF bands)
- Basis of Ham radio



HF (High Frequency) radio communication link

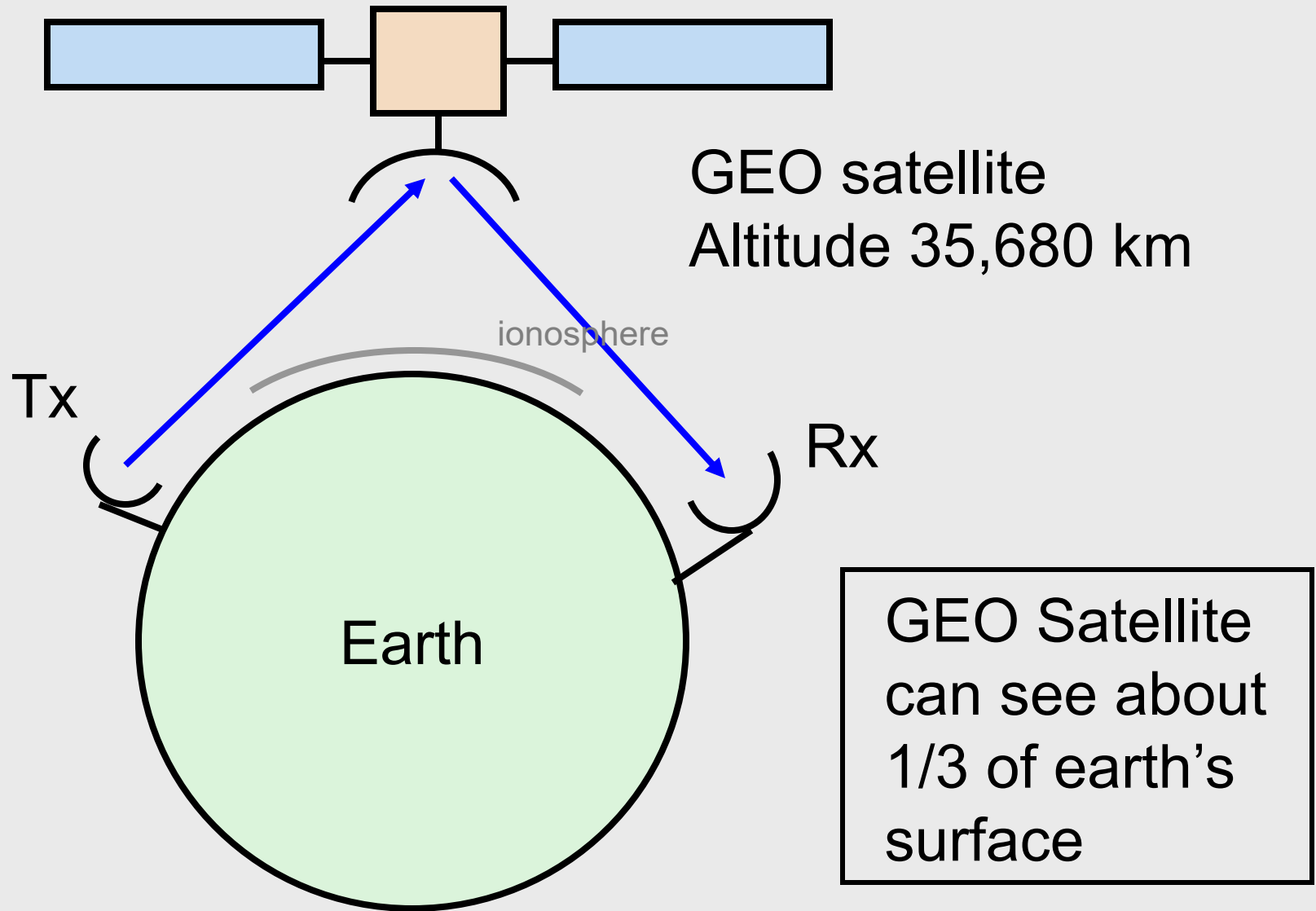
Not to scale



Long-Distance Radio Communications (3/3)

- Another solution (suggested by Arthur C. Clarke, *Wireless World* magazine) is to locate a repeater on an artificial satellite in geostationary orbit
- A very bold idea for 1945!

GEO satellite communication link



Why *Satellite* Communications?

- Large area coverage
- Links can be established quickly
- Access to remote regions (oceans, poles)
- Broadcasting – one source, many receivers

Examples:

- DBS-TV (Direct Broadcast Satellite TV)
- Video distribution for cable systems
- Internet service



Course Details

- Instructor: Dr. J. Michael Ruohoniemi
- Office: 341 Durham Hall
- Phone: (540) 231-1482
- **E-mail:** **mikeruo@vt.edu**
- Lecture time: M W 4:00 – 5:15 pm
- Lecture room: TORG 1050

Course Details

- ***IMPORTANT*** Include ECE 4644 or ECE 5610 in the Subject heading of all email!
- Tutorials will start on February 9 and run after Monday lectures (5:20 – 6:15 PM)
- Contact the instructor for office hours as needed

Course Textbook

- Class text:
Pratt and Allnutt
Satellite Communications,
John Wiley and Sons, NY, 3rd ed., 2020
ebook: ISBN 9781119482055
hardback: ISBN 9781119482178
- Text will be supplemented with additional materials made available on the class website

Course Textbook

- This week: Read *lightly* through Chapter 1
- Next week: Read through the end of Section 2.5 in Chapter 2

Class Web Site

- Canvas will be used for posting all materials in this class

What's there:

- Announcements – regular updates
- Course Information – syllabus, etc.
- Course Documents – Lecture slides, Lecture recordings, Lecture notes, review material,
- Assignments – homework and solutions

Homework Submission

- On-campus students: submit Homework as hard copy (in class, at the start of class)
- Off-campus students: submit Homework by email to mikeruo@vt.edu in PDF format (by the start of class)

Intended Audience

- ECE 4644 / ECE 5610 is intended for senior-level undergraduate students and graduate students with a good background in E/M waves, i.e., ECE 3105 and / or 3106
- The course will appeal to students interested in the wider applications of communications, radar, and radio systems
- The course material is somewhat flexible based on instructor and student interests

Schedule of Classes - 2026

- First lecture: Wednesday, January 28
- Spring Break: March 7 - 15
- Last class: Wednesday, May 6
- Reading Day: Thursday, May 7
- Final Exam: Monday, May 11, 7:45 – 9:45 AM

Grading

- Grade distribution

Homework	25%
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Exams	50%	(Mid-term, Final)
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Project	25%
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Grading

- Grading: All written work will count for grade
Part of the grade in exams and homework will be based on the quality of your writing
- Make sure that anything you write (including e-mails) is grammatically correct and properly presented

Grading

- All written submissions must be entirely your own work (except for group project work or where group homework is authorized)
- Discussion about how to go about solving HW problems is okay!
- Emailing the instructor is also okay, but make a start on the problem first
- Let me know if you are having difficulties

Honor Code (1/2)

- All students in this class are bound by the University's Honor Code
- Students shall not cheat, falsify, plagiarize, or otherwise obtain an unfair advantage in any work that is submitted for a grade
- Discussion is encouraged but not plagiarism
- Submitted work must be the student's own, and only the student's

Honor Code (2/2)

- Part II of the Syllabus provides further details
- Also see <https://www.honorsystem.vt.edu/>
- Note that for homework, undergraduate students are required to write out the following pledge:

“I have neither given nor received unauthorized assistance on this assignment”

Expected Background

- First course in communications systems and theory, e.g., ECE 3614
- Relationship between time waveforms and frequency spectra
- Fourier series and transform
- Bandwidth, filters, modulation techniques
- These topics will be reviewed as needed

Expected Background

- If you lack a background in communication systems you should:
- Talk to me
- Review posted background materials
- Learn the needed material from a basic text in communication systems

e.g. Stremler:

Introduction to Communication Systems

Roden, Couch, Lathi, Haykin

Homework

- There will be a homework assignment most weeks
- Generally issued on a Wednesday and due the following Wednesday (starting next week)
- Look at website under Files
- Please notify me if you are having difficulty in meeting a homework assignment deadline (before the deadline)

Homework # 0 – Bio Sketch

- Posted to the Canvas Files area
- Due Monday (February 2) by the start of class
- Worth 5 points – make it concise and informative
- Write a brief biographical sketch of yourself as described in the homework assignment
- Text only
- Max: 1 page

What you will learn in this course (1/2)

- How a satellite is launched and why it stays in orbit
- The systems carried by a communication satellite – transponders, antennas, power ...
- How to design and analyze a communication link
- Link budgets, system noise temperature ...

What you will learn in this course (2/2)

- The digital modulations used by satellite links
- How to share a transponder between several links – the multiple access problem
- FDMA, TDMA, CDMA, Random Access
- Error control in digital links
- Case studies
- Current developments

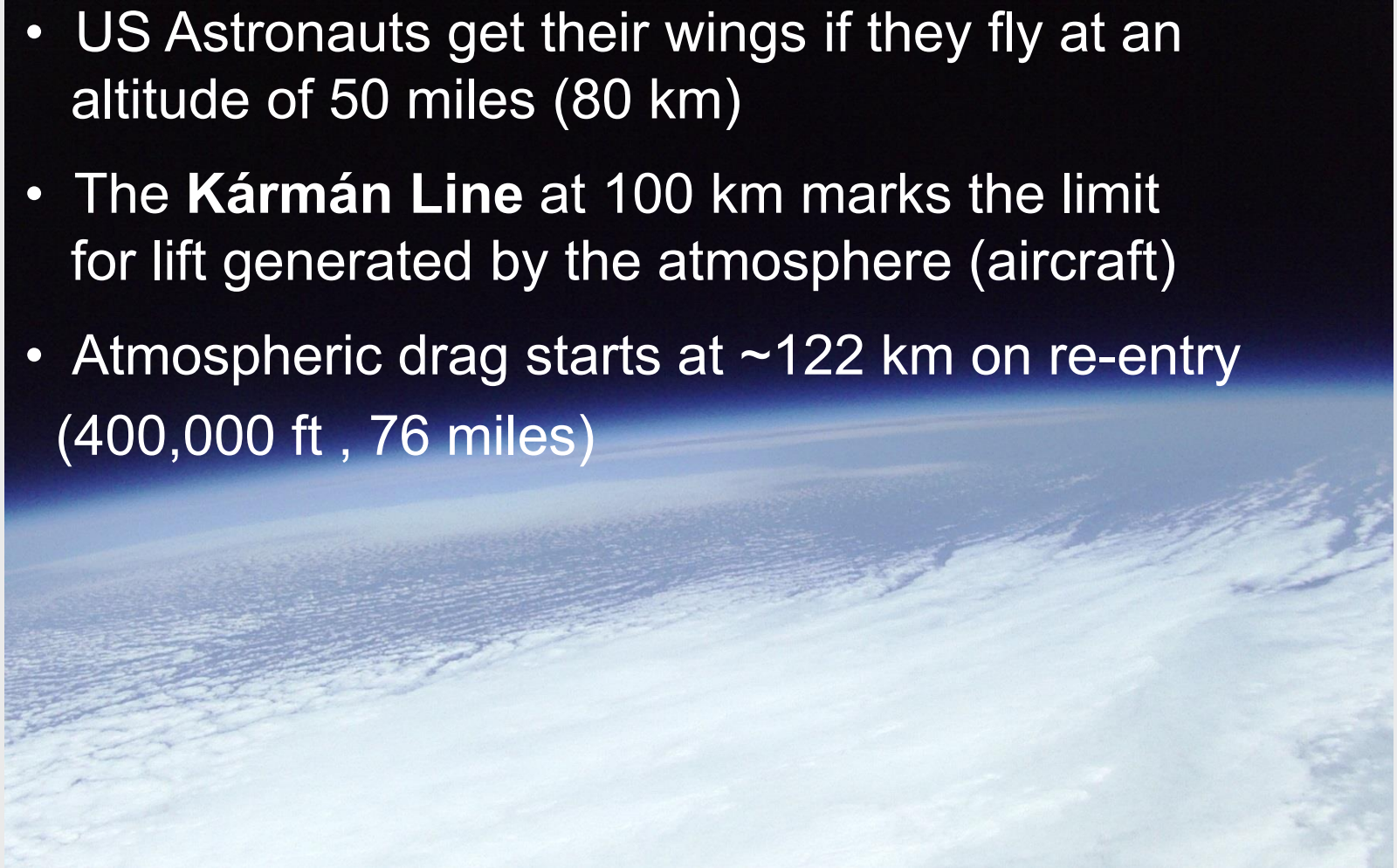
ECE 4644 / 5610 Satellite Communications

What you will be able to do by the end of this course:

- Analyze a satellite communication link to determine carrier to noise ratios and communications capacity
- Design a satellite link to meet a specification
- Design transmitting and receiving earth stations at the block diagram level

Where does Space begin?

- US Astronauts get their wings if they fly at an altitude of 50 miles (80 km)
- The **Kármán Line** at 100 km marks the limit for lift generated by the atmosphere (aircraft)
- Atmospheric drag starts at ~122 km on re-entry (400,000 ft , 76 miles)



Orbit Altitudes and Periods

- GEO: Geostationary
 - Geostationary (~24 hr period, 36,000 km over equator)
 - Geosynchronous (~24 hr inclined orbit)
- MEO: Medium Earth Orbit
 - 2000 to 36,000 km (usually 10,000 – 15,000 km)
 - GPS, Galileo, GLONASS (~12 hr period and 20,000 km)
- LEO: Low Earth Orbit
 - 250 km to 1200 km (~100 min period)
 - ISS (425 km)
 - Radiation belts above 1200 km damage electronics

Satellite Orbits

- Satellite will stay in orbit if velocity is sufficiently high
 - Orbital velocity defines altitude
 - For orbits close to earth $v \sim 7.7$ km/s (ISS)
- Orbits are categorized by the inclination of the plane of the orbit wrt the equatorial plane
 - GEO orbit has 0° inclination
 - Inclined orbit is tilted with respect to equatorial plane
 - Polar orbit (goes over the poles) has 90° inclination

Satellite Orbits

- LEO orbits have periods of 92 - 110 minutes
- Satellite is visible above the horizon for an observer for only 10 - 15 minutes
- *A constellation* of many LEO satellites is required for global coverage
- LEO orbits are used for:
 - Earth surveillance
 - Satellite phones (Iridium, Globalstar)
 - Internet access (Starlink, OneWeb)

Iridium Global Constellation



LEO orbit

66 sats

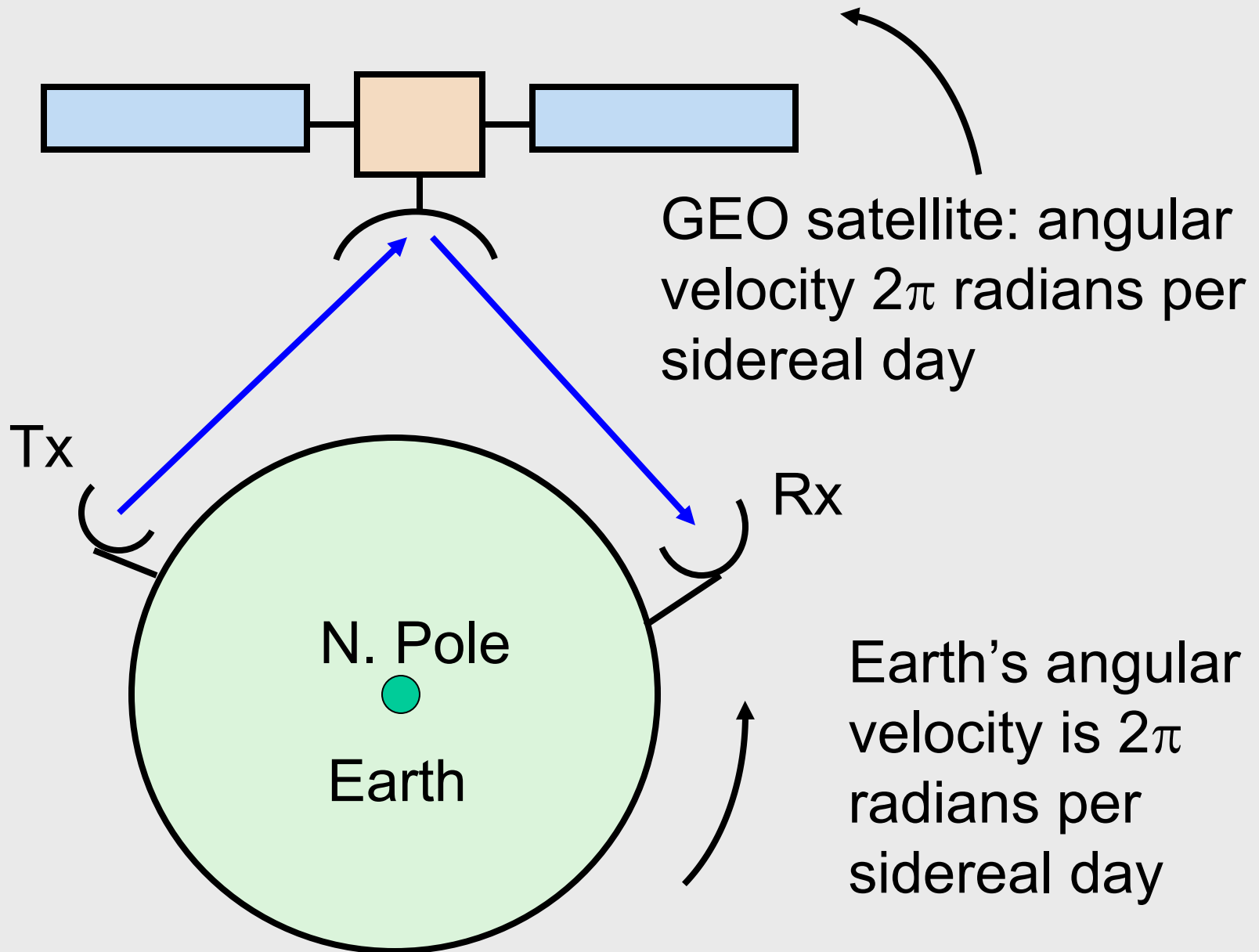
$h = 780$
km

$i = 86.4^\circ$

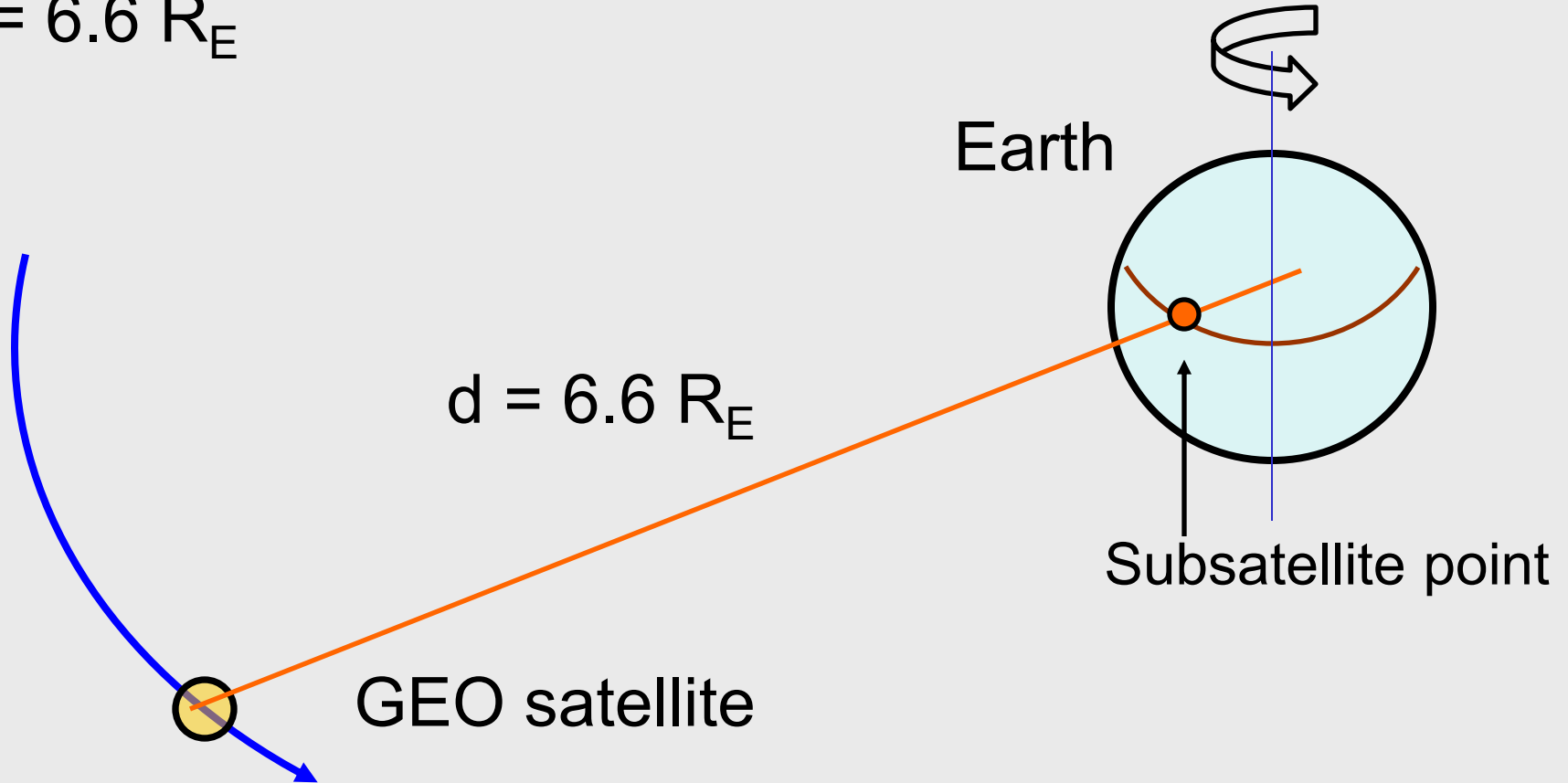
Six orbital
planes

Satellite Orbits

- The highest value *individual* satellites have been flown in *Geostationary Earth Orbit* (GEO)
- GEO satellites
 - are above the equator
 - have angular velocity equal to earth's angular velocity
 - appear stationary above a point on the equator



GEO satellites are located at a distance from the center of the Earth = $6.6 \times$ radius of Earth, i.e.,
 $d = 6.6 R_E$



Almost to scale

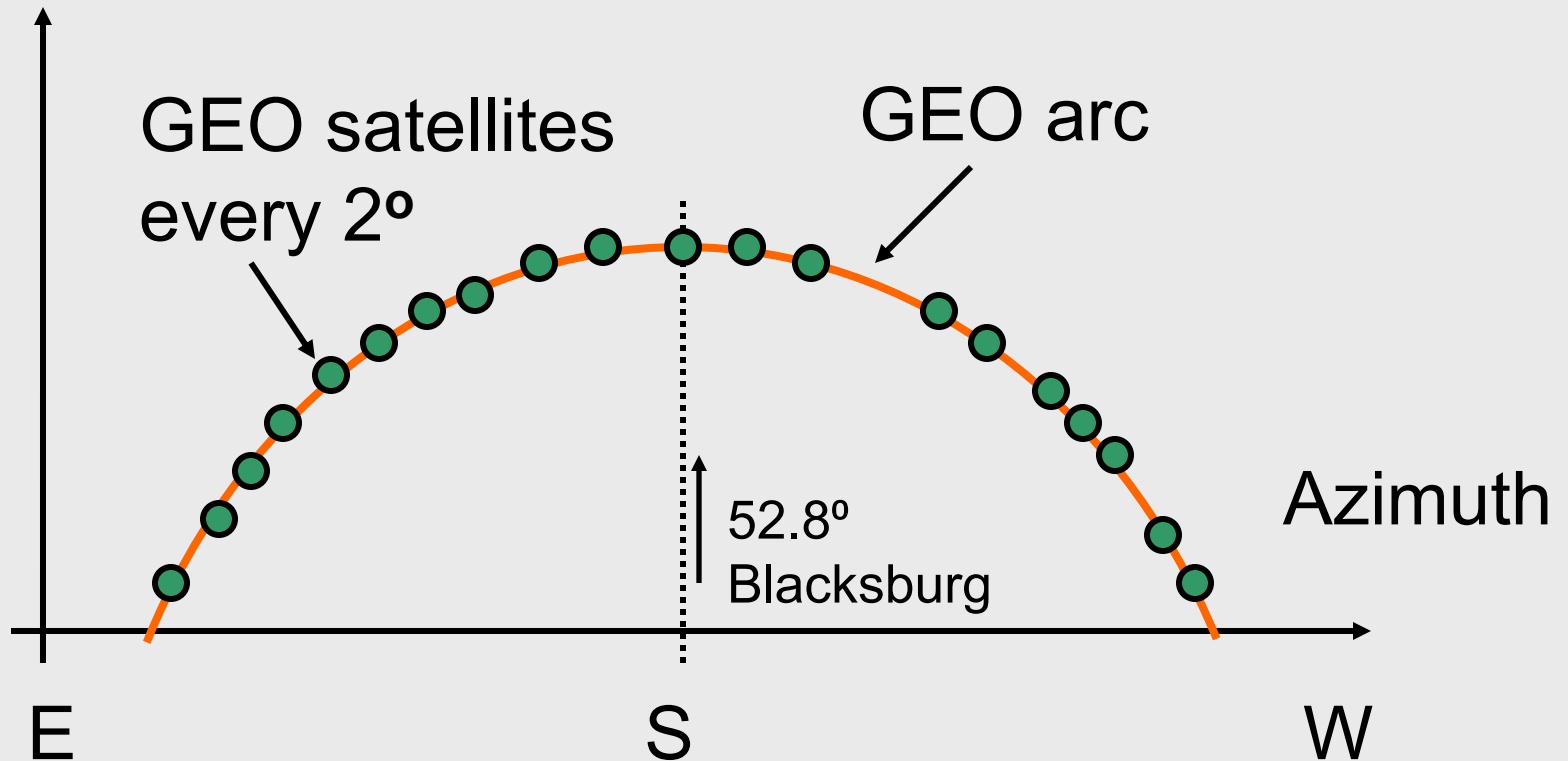
GEO Orbit

- In equatorial plane, circular
- Orbital Period = 23 h 56 min 4.091 s
= one *sidereal day*
- Earth makes one complete rotation with respect to the distant stars in one sidereal day
- Earth makes one complete rotation with respect to the sun in one *solar day* (24 hr)
- The difference of 3 min 56 s is due to Earth's movement in its orbit around the sun

GEO Satellites

- GEO satellites appear stationary in the sky
- Travel at 3 km per second
- From an *earth station* GEO satellites appear to be distributed along an arc in the sky
- This arc is located directly above Earth's equator
- From Blacksburg (37.2° N) the peak elevation angle of the arc is 52.8° (due south)

Elevation angle



GEO arc as seen from northern hemisphere
“Clarke belt”

GEO Satellites

- GEO satellites are large, heavy, and expensive
- Typical weight: 6,000 kg
- Typical size: 30 m (100 ft) over solar wings
- Typical cost: \$125 M
- Typical launch: \$50 M
- About 2% of launch weight of rocket (ELV) reaches GEO orbit
- 4000 kg satellite needs 200,000 kg launcher

GEO Satellites

- Cost of building and launching GEO satellites must be recovered by selling communications capacity
- Typical cost of satellite and launch: \$150 M
- Typical lifetime: 15 years
- Operating costs: \$10 M per year
- Earnings must exceed \$20 M per year to pay back loans and interest
- Sell 20 transponders (repeaters) at \$1M each per year

1st Lecture Summary (1/2)

- Satellite communications is a well-established technology (~ 60 years old)
- Commercial use began in 1965 with *Early Bird*
- Satellite orbits are characterized by altitude as LEO, MEO, and GEO and also by inclination
- Orbital altitude, velocity, and period are closely related

Summary (2/2)

- GEO satellites have dominated until recently:
 - high capacity, low cost per transmitted bit
 - direct broadcast TV is major application
- LEO satellite systems are challenging this dominance with large fleets
 - telephone and data service
 - global broadband
- The field of satcom is developing rapidly with significant developments almost every month!

